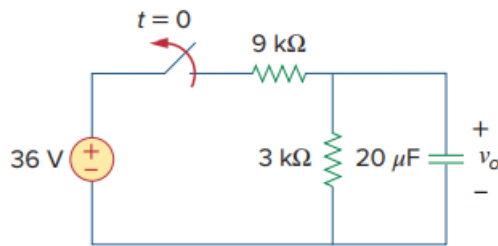


Problem

For the circuit in Fig. 7.90, find $v_o(t)$ for $t > 0$. Determine the time necessary for the capacitor voltage to decay to one-third of its value at $t = 0$.

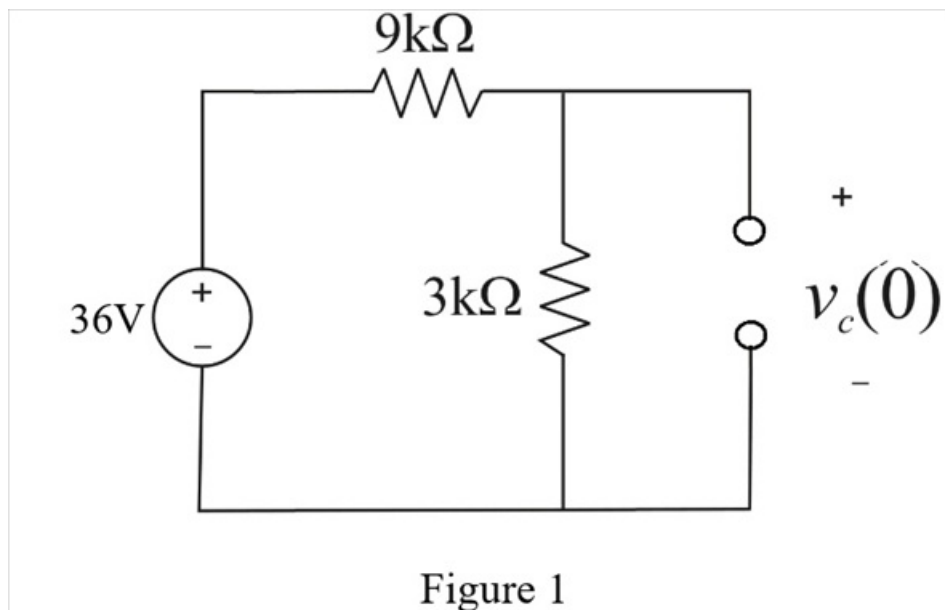
Fig. 7.90:



Step-by-step solution

Step 1 of 5

For $t < 0$ switch is closed and the capacitor acts like an open circuit for dc.



Step 2 of 5

Apply the voltage division rule.

$$v_c(0) = 36 \times \left(\frac{3 \times 10^3}{9 \times 10^3 + 3 \times 10^3} \right)$$

$$v_c(0) = 9V$$

Step 3 of 5

For $t > 0$ the switch is opened and we have the RC circuit shown in Figure 2.

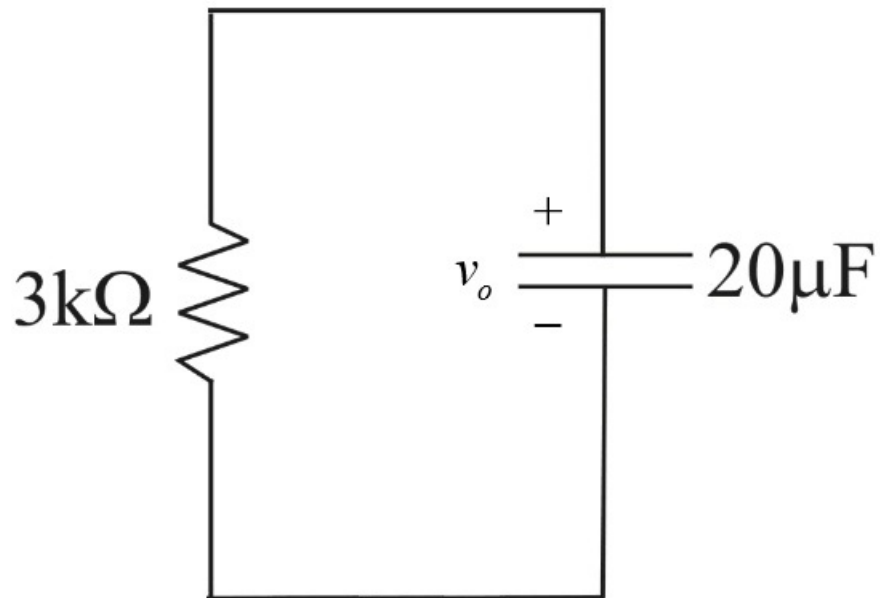


Figure 2

[Comments \(2\)](#)



Anonymous

Why is the capacitor not affecting the 9k ohm resistor?



Anonymous

switch is opened, 36v and 9k are connected to open circuit, no effect

Step 4 of 5

Calculate the time constant.

$$\tau = RC$$

$$\tau = 3 \times 10^3 \times 20 \times 10^{-6}$$

$$\tau = 60 \text{ m sec}$$

Thus, the voltage across capacitor is,

$$v_o(t) = v_c(0) e^{\frac{-t}{\tau}} \text{ V } t > 0$$

$$v_o(t) = 9 e^{\frac{-t}{60 \times 10^{-3}}} \text{ V for } t > 0$$

Step 5 of 5

Capacitor voltage at $t = 0$ is $v_o(0) = 9V$.

Let the time required to voltage to decay to one-third of its value at $t = 0$ is t_1 ,

$$v_o(t_1) = \frac{v_o(0)}{3}$$

$$v_o(t_1) = \frac{9}{3}$$

$$v_o(t_1) = 3V$$

$$3 = 9e^{\frac{-t}{60 \times 10^{-3}}}$$

$$\frac{1}{3} = e^{\frac{-t}{60 \times 10^{-3}}}$$

$$\frac{-t}{60 \times 10^{-3}} = \ln\left(\frac{1}{3}\right)$$

$$t = 0.0659 \text{ sec}$$

Hence, the time required is 0.066 sec.